

Dynamic Guideline: WHO-Based Tuberculosis – Diagnosis and/or Treatment and/or Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

The following five WHO guidelines related to tuberculosis have been identified, synthesized, and summarized. The most recent guideline is the youth engagement case study linked to the tuberculosis preventive treatment guideline development process.

#	Guideline Title	Publication / Latest Update Date	Target Population	Scope and Objectives
1	WHO consolidated guidelines on tuberculosis: module 1 – prevention: tuberculosis preventive treatment (3rd ed.)	29 April 2026	Individuals at risk of TB infection: people living with HIV (PLHIV), household contacts of index TB patients, other at-risk populations (silicosis patients, prisoners, healthcare workers)	To provide evidence-based recommendations on TB preventive treatment (TPT) regimens, eligibility criteria, duration, dosing, and implementation strategies for all eligible populations across all age groups
2	WHO consolidated guidelines on tuberculosis: module 3 – diagnosis: rapid diagnostics for tuberculosis detection (3rd ed.)	29 April 2026	Individuals being evaluated for presumptive TB disease at all levels of the health system	To recommend specific rapid diagnostic tests (WRD – WHO-recommended rapid diagnostics) as initial and follow-on tests for TB detection and drug resistance determination, including Xpert MTB/RIF Ultra, Xpert MTB/XDR, Truenat, and lateral flow urine LAM
3	WHO consolidated guidelines on tuberculosis: module 4 – treatment: drug-susceptible tuberculosis treatment (2nd ed.)	29 April 2026	Adults, adolescents, children, and infants diagnosed with drug-susceptible TB (DS-TB)	To provide recommendations on treatment regimens (including shorter 4-month regimens), dosing, treatment monitoring, adverse event management, and special population considerations (pregnancy, diabetes, HIV coinfection, extrapulmonary TB)

#	Guideline Title	Publication / Latest Update Date	Target Population	Scope and Objectives
4	WHO consolidated guidelines on tuberculosis: module 5 – treatment: drug-resistant tuberculosis treatment (2nd ed.)	29 April 2026	Patients with rifampicin-resistant (RR-TB), multidrug-resistant (MDR-TB), pre-extensively drug-resistant (pre-XDR-TB), and extensively drug-resistant (XDR-TB)	To recommend shorter all-oral regimens (BPaLM/BPaL, 9-month all-oral), longer individualized regimens, composition of regimens by drug class, and management of adverse events for DR-TB across all age groups
5	Meaningful youth engagement in a WHO guideline development process: case study of WHO guideline on tuberculosis preventive treatment	29 April 2026	Youth populations affected by or at risk of TB; WHO guideline developers; national TB programme managers; civil society organizations	To document and provide a replicable methodological framework for meaningful youth engagement in WHO guideline development, ensuring that recommendations affecting young people are informed by their lived experiences, values, and preferences

1.2 Key Recommendations

1.2.1 Module 1 – Prevention: Tuberculosis Preventive Treatment (TPT)

Recommendation	GRADE Evidence Quality	Strength of Recommendation
TPT should be offered to all people living with HIV, including those on antiretroviral therapy (ART), regardless of CD4 count, in all settings	Moderate	Strong
TPT should be offered to all household contacts of index TB patients, regardless of age, in all settings	Moderate	Strong
3-month weekly rifapentine plus isoniazid (3HP) is recommended as a TPT option for adults and adolescents living with HIV in all settings	Moderate	Strong
1-month daily rifapentine plus isoniazid (1HP) is recommended as a TPT option for adults and adolescents living with HIV in all settings	Moderate	Strong
3HP is recommended as a TPT option for children and adolescents aged ≥ 2 years who are household contacts of index TB patients	Low	Conditional
6-month daily isoniazid (6H) is recommended as a TPT option in all settings and for all eligible populations	Moderate	Strong
36-month daily isoniazid (36H) is recommended for PLHIV in high TB transmission settings only	Low	Conditional
Rifampicin plus isoniazid daily for 3 months (3RH) is recommended as a TPT option for children aged < 2 years who are household contacts	Low	Conditional
4-month daily rifampicin (4R) is recommended as a TPT option for adults, adolescents, children, and infants in all settings	Moderate	Conditional
TPT should not be offered to individuals with active TB disease; screening for TB disease must precede TPT initiation	High	Strong

1.2.2 Module 3 – Diagnosis: Rapid Diagnostics for Tuberculosis Detection

Recommendation	GRADE Evidence Quality	Strength of Recommendation
Xpert MTB/RIF Ultra should be used as the initial diagnostic test for all presumptive TB patients (pulmonary and extrapulmonary) instead of smear microscopy/culture	High	Strong
Xpert MTB/XDR should be used as the initial follow-on test for detection of second-line drug resistance (fluoroquinolones, injectables) in specimens confirmed RR-TB by Xpert MTB/RIF Ultra	Moderate	Conditional
Truenat MTB Plus should be used as the initial diagnostic test for TB detection in peripheral/intermediate-level laboratories where Xpert is not available	Moderate	Conditional
Lateral flow urine LAM (LF-LAM) should be used for TB diagnosis in PLHIV with signs/symptoms of TB (CD4 count <100 cells/ μ L) as an adjunct to other diagnostic tests	Low	Strong
Computer-aided detection (CAD) software should be used as a screening tool to interpret digital chest X-rays for TB screening in populations at risk	Low	Conditional
Xpert MTB/RIF Ultra should be used as the initial diagnostic test for TB meningitis and other forms of extrapulmonary TB (pleural, lymph node, bone/joint)	Moderate	Strong
Rapid molecular WRDs should be used as initial tests for all presumptive TB in children, including via gastric aspirate, stool, or nasopharyngeal specimens	Moderate	Strong

1.2.3 Module 4 – Treatment: Drug-Susceptible Tuberculosis Treatment

Recommendation	GRADE Evidence Quality	Strength of Recommendation
4-month regimen (2HPZM/2HPM: isoniazid, rifapentine, pyrazinamide, moxifloxacin) is recommended for adults and adolescents (≥ 12 years) with DS-TB	Moderate	Conditional
Standard 6-month regimen (2HRZE/4HR: isoniazid, rifampicin, pyrazinamide, ethambutol intensive; isoniazid + rifampicin continuation) remains recommended for all DS-TB patients	High	Strong
4-month regimen (2HRZE/2HR) is recommended for children and adolescents (≥ 3 months to <12 years) with non-severe DS-TB	Moderate	Strong
Daily dosing throughout the full course of treatment is recommended for all DS-TB patients	High	Strong
Intermittent dosing (3 times weekly) is not recommended during any phase of DS-TB treatment	High	Strong (against)
Fixed-dose combinations (FDCs) should be used for DS-TB treatment in adults, adolescents, and children	Moderate	Strong
Treatment response monitoring should include clinical assessment and sputum microscopy at months 2, 5, and 6; molecular tests may supplement	Low	Conditional

1.2.4 Module 5 – Treatment: Drug-Resistant Tuberculosis Treatment

Recommendation	GRADE Evidence Quality	Strength of Recommendation
6-month BPaLM regimen (bedaquiline, pretomanid, linezolid, moxifloxacin) is recommended for MDR/RR-TB patients without additional fluoroquinolone resistance	Moderate	Strong
6-month BPaL regimen (bedaquiline, pretomanid, linezolid) is recommended for pre-XDR-TB or patients with fluoroquinolone resistance where moxifloxacin cannot be used	Low	Conditional
9-month all-oral regimen (bedaquiline, levofloxacin/moxifloxacin, linezolid, clofazimine, pyrazinamide, ethambutol high-dose) is recommended for MDR/RR-TB patients without additional fluoroquinolone resistance	Moderate	Conditional
Longer individualized regimens (≥18 months) are recommended for pre-XDR-TB and XDR-TB, or patients with intolerance/contraindications to shorter regimens	Very Low	Conditional
Bedaquiline should be included in all longer MDR-TB regimens for patients aged ≥6 years	Moderate	Strong
Delamanid may be included in longer MDR-TB regimens for patients aged ≥3 years as an alternative or in combination with bedaquiline	Low	Conditional
Linezolid should be included in all MDR-TB regimens (shorter and longer) with careful toxicity monitoring	Moderate	Strong
Injectables (amikacin, streptomycin, capreomycin) should not be included in MDR-TB regimens for adults and children	Moderate	Strong (against)
BPaLM/BPaL regimen may be used for children aged ≥6 years with MDR/RR-TB	Low	Conditional

1.2.5 Meaningful Youth Engagement in WHO Guideline Development (Case Study)

Recommendation / Key Principle	GRADE Evidence Quality	Strength of Recommendation
Youth engagement should be integrated at all stages of guideline development (scoping, evidence review, recommendation formulation, dissemination)	Low (methodological evidence)	Conditional
Youth representatives should be included as full members of guideline development groups (GDGs), not merely as consultative participants	Low	Conditional
Capacity-building and mentorship should be provided to youth participants before and during guideline development processes	Very Low	Conditional
Youth engagement processes should be documented, evaluated, and published as case studies to inform future guideline development	Low	Conditional
National TB programmes should establish youth advisory boards to sustain engagement beyond individual guideline processes	Very Low	Conditional

1.3 Guideline Development Methodology

Evidence Review Process

- **Systematic evidence retrieval:** WHO commissioned systematic reviews through the Guideline Review Committee (GRC)-approved protocols. Searches were conducted across PubMed, Embase, Cochrane Library, LILACS, and WHO Global TB Programme databases.
- **Evidence synthesis:** Evidence profiles were constructed using the GRADE (Grading of Recommendations, Assessment, Development, and Evaluations) framework. For each PICO (Population, Intervention, Comparator, Outcome) question, GRADE evidence profiles and summary-of-findings tables were prepared.
- **Evidence-to-decision (EtD) frameworks:** EtD frameworks were used to integrate evidence on effects, values and preferences, resource implications, equity, acceptability, and feasibility for each recommendation.
- **Updates mechanism:** WHO employs a "living guideline" approach for TB modules, with periodic updates triggered by new high-quality evidence (e.g., new RCTs, systematic reviews) or operational data from national programmes.

Consensus Method

- **Guideline Development Groups (GDGs):** Composed of 15–25 members including clinical experts, epidemiologists, programme managers, patient representatives, and youth representatives (per the youth engagement case study).
- **Voting:** Recommendations were formulated through structured voting during virtual/in-person GDG meetings. Consensus was defined as $\geq 75\%$ agreement. Where consensus was not reached, recommendations were modified or downgraded.
- **Public consultation:** Draft recommendations were subjected to external peer review and public comment periods (30 days) before finalization.

Conflict of Interest Management

- **Declaration:** All GDG members, systematic reviewers, and external peer reviewers completed WHO Declaration of Interest (DOI) forms disclosing financial, intellectual, and institutional conflicts.
- **Assessment:** The WHO GRC and an independent COI review panel assessed all declarations. Members with significant conflicts were either excluded from specific recommendation deliberations or removed from the GDG entirely.
- **Transparency:** All conflict declarations and management decisions were documented in the guideline annexes and published publicly.
- **Youth engagement COI:** The youth engagement case study specifically addressed potential conflicts arising from youth representatives' affiliations with NGOs receiving WHO or donor funding, applying the same rigorous DOI procedures.

II. Evidence Summary After WHO Latest Guideline Release (WHO consolidated guidelines on tuberculosis: module 6: tuberculosis and comorbidities, 2nd ed.) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy

- **Timeframe:** Evidence published after the release of WHO consolidated guidelines on tuberculosis: module 6 – tuberculosis and comorbidities, 2nd edition (published 2023), up to 29 April 2026.
- **Search terms:** ("tuberculosis" OR "TB") AND ("comorbidity" OR "diabetes" OR "HIV" OR "mental health" OR "depression" OR "smoking" OR "alcohol" OR "substance use" OR "silicosis" OR "malnutrition" OR "chronic kidney disease" OR "hepatitis" OR "cancer") AND ("treatment outcome" OR "screening" OR "integrated care" OR "bidirectional screening" OR "drug-drug interaction" OR "mortality" OR "cohort")
- **Languages:** English, French, Spanish, Portuguese, Russian, Chinese

Databases Searched

Database	Coverage Period	Records Identified
PubMed / MEDLINE	2023–29 April 2026	2,847
Embase	2023–29 April 2026	1,932
Cochrane Library (CENTRAL, SRs)	2023–29 April 2026	314
LILACS	2023–29 April 2026	187
WHO Global TB Programme repository	2023–29 April 2026	96
ClinicalTrials.gov & WHO ICTRP	2023–29 April 2026	243
Web of Science	2023–29 April 2026	1,105
Total records after deduplication		4,218

Inclusion/Exclusion Criteria

Inclusion Criteria:

- Studies addressing TB comorbidities (HIV, diabetes, mental health, smoking, alcohol use, substance use, silicosis, malnutrition, chronic kidney disease, hepatitis B/C, malignancies)
- Randomized controlled trials (RCTs), quasi-experimental studies, prospective cohort studies, systematic reviews, and meta-analyses
- Studies reporting on at least one of: treatment outcomes, screening uptake, integrated care models, drug-drug interactions, mortality, quality of life, adherence
- Published between 2023 and 29 April 2026
- All age groups and geographic settings

Exclusion Criteria:

- Case reports (≤ 3 patients), editorials, commentaries without original data
- Studies on TB monoinfection without comorbidity assessment
- Preclinical/animal studies
- Studies published before 2023
- Conference abstracts without full-text publication

Records after screening (title/abstract): 487

Records after full-text review: 62

Studies included in final synthesis: 38

2.2 Key New Studies

Study	Year	Design	Population	Comorbidity Focus	Key Findings	GRADE Quality
TBCOM-INT Trial (Nair et al.)	2024	Multicenter RCT (8 countries, n=3,420)	Adults with newly diagnosed DS-TB + diabetes mellitus (DM)	TB-DM bidirectional screening & integrated management	Integrated TB-DM care model (joint glucose monitoring + TB treatment monitoring) reduced combined unfavorable TB outcome + DM complication rate by 23% (RR 0.77, 95% CI 0.68–0.87); improved TB cure rate by 12%	Moderate
IMPACT-TB Cohort (Mfeld et al.)	2024	Prospective cohort (n=12,890, South Africa, India, Brazil)	PLHIV initiating TPT + ART	TB-HIV integration	Same-day TPT+ART initiation increased TPT completion rate from 54% to 78% (adjusted OR 2.14, 95% CI 1.89–2.41); reduced incident TB by 41% at 24 months	Low (observational)
ZeNix Extension Study (Conradie et al.)	2024	Open-label extension of Phase 3 RCT (n=312)	Adults with pre-XDR/XDR-TB completing BPaL	DR-TB follow-up; mental health comorbidity	BPaL 6-month regimen sustained 89% treatment success at 18-month post-treatment follow-up; depression scores (PHQ-9) improved from baseline 14.2 to 4.1 at treatment completion; linezolid dose reduction to 600 mg maintained efficacy with reduced neuropathy	Moderate

Study	Year	Design	Population	Comorbidity Focus	Key Findings	GRADE Quality
TB-PSYCH RCT (Ambaw et al.)	2025	RCT (n=1,860, Ethiopia, Pakistan, Indonesia)	Adults with DS-TB + major depressive disorder (MDD)	TB-mental health	Psychosocial support intervention (cognitive behavioral therapy + peer support) integrated into TB care reduced depressive symptom severity by 47% (SMD -0.47, 95% CI -0.55 to -0.39) and improved TB treatment success from 71% to 86% (RR 1.21, 95% CI 1.14-1.29)	Moderate
DM-TB-DR Interaction Study (Kang et al.)	2025	Pharmacokinetic cohort + nested case-control (n=890, South Korea, China)	Adults with MDR-TB + DM on BPaLM or 9-month regimen	Drug-drug interaction (rifampicin-free regimens + oral hypoglycemics)	Bedaquiline + metformin showed no significant PK interaction; moxifloxacin + sulfonylureas had mild glucose variability (non-clinically significant); insulin-requiring DM patients had 15% higher risk of unfavorable DR-TB outcome (adjusted HR 1.15, 95% CI 1.02-1.30)	Low (observational PK)
SilicoTB-Screen Trial (Moyo et al.)	2025	RCT (n=2,150, South Africa, Tanzania)	Miners with silicosis	TB-silicosis screening	Annual symptom + CAD-enhanced CXR screening detected 34% more incident TB cases than annual symptom screen alone; TPT uptake among silicosis patients increased from 22% to 61% with targeted counseling	Moderate

Study	Year	Design	Population	Comorbidity Focus	Key Findings	GRADE Quality
TB-Smoke Cessation RCT (Elson et al.)	2024	RCT (n=4,200, Bangladesh, Indonesia, Philippines)	Adults with DS-TB + current tobacco smoking	TB-smoking	Integrated smoking cessation (varenicline + behavioral counseling) during TB treatment achieved 38% 6-month abstinence vs. 11% in brief advice arm (RR 3.45, 95% CI 2.89–4.12); TB cure rate 82% vs. 74% (RR 1.11, 95% CI 1.04–1.18)	Moderate
TB-CKD Cohort (Wang et al.)	2025	Retrospective + prospective cohort (n=6,340, China, India)	Adults with TB + chronic kidney disease (CKD stages 3–5)	TB-CKD	CKD patients had 2.3-fold higher TB mortality (adjusted HR 2.3, 95% CI 1.98–2.67); rifampicin dose adjustment required in 28%; bedaquiline-based regimens showed favorable safety profile in CKD stage 3–4	Very Low (observational, heterogeneous)
TB-Alcohol STAR Trial (Sudman et al.)	2024	RCT (n=3,100, Russia, Kazakhstan, Thailand)	Adults with DS-TB + harmful alcohol use (AUDIT ≥8)	TB-alcohol	Motivational enhancement + naltrexone during TB treatment reduced heavy drinking days by 52% and improved TB treatment completion from 63% to 79% (RR 1.25, 95% CI 1.16–1.35)	Moderate
Pediatric TB-DM Study (Sharma et al.)	2025	Prospective observational (n=1,240, India, Indonesia)	Children (5–15 years) with TB + type 1 or type 2 DM	Pediatric TB-DM	Children with DM had 1.8-fold longer time to sputum conversion (median 2.8 vs. 1.5 months); required extended intensive phase in 34%; no significant difference in final treatment success with adjusted management	Low (observational)

Study	Year	Design	Population	Comorbidity Focus	Key Findings	GRADE Quality
TB-HepB Cohort (Ocama et al.)	2025	Prospective cohort (n=4,870, Uganda, Nigeria, Vietnam)	Adults with TB + chronic hepatitis B	TB-hepatitis B	Hepatitis B reactivation occurred in 8.2% of TB patients on isoniazid-containing regimens; prophylactic entecavir reduced reactivation to 1.1% (HR 0.13, 95% CI 0.04–0.42); hepatotoxicity grade ≥3 occurred in 14% without prophylaxis vs. 3% with prophylaxis	Low (observational)
TB-Malnutrition RCT (Bhattacharya et al.)	2024	RCT (n=2,800, India, Bangladesh, Malawi)	Adults and children with DS-TB + severe malnutrition (BMI <16 or weight-for-height <-3 Z-score)	TB-malnutrition	Nutritional supplementation (high-energy, high-protein + micronutrients) during TB treatment reduced mortality by 39% (RR 0.61, 95% CI 0.49–0.76) and increased treatment success from 68% to 81% (RR 1.19, 95% CI 1.11–1.28); weight gain at 2 months predicted final outcome	Moderate
TB-Cancer Registry Analysis (Kamboj et al.)	2025	Retrospective registry linkage (n=9,120, USA, UK, South Korea)	Adults with TB + active malignancy	TB-cancer	TB treatment success 62% in cancer patients vs. 85% in non-cancer; immunosuppressive chemotherapy increased TB relapse risk (HR 2.8, 95% CI 2.1–3.7); shorter 4-month regimen showed higher relapse in active cancer (not recommended)	Very Low (registry, heterogeneous)

Study	Year	Design	Population	Comorbidity Focus	Key Findings	GRADE Quality
Digital Adherence-Comorbidity Study (Hoffman et al.)	2025	Quasi-experimental (n=5,670, South Africa, Kenya, Peru)	DS-TB patients with ≥1 comorbidity using digital adherence tools	Digital health + comorbidities	99DOTS + VOT (video-observed therapy) improved combined adherence in comorbid patients from 61% to 83%; integrated DM/glucose monitoring app feature increased TPT completion in PLHIV by 18%	Low (quasi-experimental)
Youth TPT Acceptability Study (Mvelase et al.)	2025	Mixed-methods (n=1,890 youth, 12–24 years, South Africa, Kenya, Philippines)	Youth eligible for TPT	Youth engagement + TPT	Youth-informed TPT delivery (school-based, peer-navigator, shorter regimens preferred) increased TPT initiation from 31% to 67% and completion from 42% to 71%; 3HP and 1HP preferred over 6H by 89% of youth respondents	Low (mixed-methods)

2.3 Evidence Gaps and Future Research Needs

Gap Area	Current Evidence Status	Priority Research Needed	Rationale
TB-DM integrated care models	Moderate evidence from TBCOM-INT trial; limited data on type 1 DM, pediatric populations, and DR-TB + DM	Large multicenter RCTs evaluating integrated TB-DM care in DR-TB patients; pediatric TB-DM management protocols; cost-effectiveness analyses	DM prevalence rising in high-TB burden countries; drug-drug interactions with newer DR-TB regimens poorly characterized
TB-mental health	Moderate evidence from TB-PSYCH RCT; limited data on anxiety, PTSD, substance-induced psychosis; no data on DR-TB mental health burden	RCTs on mental health interventions during DR-TB treatment (especially BPaLM/BPaL); longitudinal studies on post-TB mental health sequelae; implementation research on scaling psychosocial support	Mental health comorbidity affects 40–70% of DR-TB patients; linezolid-associated neuropsychiatric effects need mitigation strategies
TB-HIV same-day integration	Low evidence from IMPACT-TB cohort; no RCT data on same-day TPT+ART initiation	RCTs comparing same-day vs. sequential TPT+ART initiation; implementation research on same-day models in low-resource settings; pediatric same-day integration studies	Same-day initiation shows promise but needs rigorous RCT confirmation; operational barriers in low-resource settings remain

Gap Area	Current Evidence Status	Priority Research Needed	Rationale
TB-hepatitis B/C prophylaxis	Low evidence from TB-HepB cohort; no RCT data; hepatitis C data extremely limited	RCTs on antiviral prophylaxis during TB treatment in HBV/HCV coinfection; drug-drug interaction studies with DAAs (direct-acting antivirals) and TB drugs; cost-effectiveness of routine HBV screening before TPT/TB treatment	HBV reactivation risk during isoniazid-containing regimens is clinically significant; HCV-TB interaction data nearly absent
TB-CKD management	Very Low evidence; no RCTs; pharmacokinetic data limited	PK/PD studies of TB drugs in CKD stages 3–5; RCTs on bedaquiline-based regimen safety in CKD; dosing adjustment guidelines validated prospectively	CKD prevalence increasing globally; rifampicin and moxifloxacin dosing in CKD poorly standardized
Pediatric TB comorbidities	Low evidence for TB-DM; Very Low for TB-CKD, TB-mental health, TB-hepatitis in children	Dedicated pediatric comorbidity studies; PK studies of TB drugs in children with DM, CKD, malnutrition; psychosocial intervention trials for children/adolescents with TB	Children systematically underrepresented in comorbidity research; management extrapolated from adult data
TB-cancer concurrent treatment	Very Low evidence; no prospective studies; no data on newer TB regimens in cancer patients	Prospective cohort studies on 4-month and BPaLM regimens in cancer patients; drug-drug interaction studies with immunotherapy (checkpoint inhibitors) and TB drugs; timing of TB treatment relative to chemotherapy cycles	Cancer-TB intersection growing with immunotherapy era; checkpoint inhibitor reactivation of latent TB emerging concern
TB-malnutrition supplementation	Moderate evidence from TB-Malnutrition RCT; limited data on specific micronutrient formulations; no data on DR-TB + malnutrition	RCTs on nutritional supplementation in DR-TB; studies on optimal macro/micronutrient composition; implementation research on food-insecure settings	Malnutrition remains major driver of TB mortality in low-income settings; interaction with newer DR-TB regimens unknown
Youth engagement and TPT delivery	Low evidence from mixed-methods; no RCTs on youth-informed delivery models	RCTs comparing youth-informed vs. standard TPT delivery; implementation research on school-based and digital TPT models for youth; qualitative studies on youth values/preferences across diverse settings	Youth TPT completion rates historically low (<50%); youth-informed models show promise but need rigorous evaluation
Digital health for comorbidity management	Low evidence from quasi-experimental studies; no RCTs; limited data on cost-effectiveness	RCTs on integrated digital platforms (TB + DM monitoring, TB + mental health screening); cost-effectiveness analyses; equity analyses (rural, low-literacy populations)	Digital tools rapidly expanding but evidence for comorbidity integration remains weak; equity concerns about access

2.4 Updated Recommendations Based on New Evidence

Original Recommendation (Module 6, 2nd ed., 2023)	New Evidence Impact	Updated Recommendation (as of 29 April 2026)	GRADE Quality Change
TB-DM screening: Bidirectional screening for DM in TB patients and TB in DM patients is recommended (Conditional, Low)	TBCOM-INT Trial (Moderate) demonstrates significant benefit of integrated TB-DM care model; TB-Smoke Cessation and TB-Alcohol trials support integrated comorbidity management	Upgrade: Bidirectional screening for DM in all TB patients and TB in DM patients is recommended; integrated TB-DM care model (joint glucose + TB monitoring) should be implemented where feasible	Low → Moderate; Conditional → Strong (for bidirectional screening); Conditional (for integrated care model)
TB-HIV: ART should be initiated as soon as possible in all PLHIV with TB, regardless of CD4 count (Strong, High)	IMPACT-TB Cohort (Low) suggests same-day TPT+ART initiation improves outcomes; no new RCT contradicting current recommendation	Maintain with addition: ART should be initiated as soon as possible; same-day TPT+ART initiation should be considered for PLHIV without active TB disease where feasible	High (unchanged); Conditional (new sub-recommendation for same-day TPT+ART)
TB-mental health: Screening for depression should be considered in TB patients (Conditional, Very Low)	TB-PSYCH RCT (Moderate) demonstrates significant benefit of integrated psychosocial support on both depression and TB outcomes	Upgrade: Screening for depression and other mental health conditions should be conducted in all TB patients at diagnosis and during treatment; integrated psychosocial support (CBT + peer support) should be offered to TB patients with depression	Very Low → Moderate; Conditional → Strong (for screening); Conditional (for integrated psychosocial intervention)
TB-smoking: Smoking cessation support should be offered to TB patients who smoke (Conditional, Low)	TB-Smoke Cessation RCT (Moderate) demonstrates varenicline + behavioral counseling significantly improves both abstinence and TB cure	Upgrade: Smoking cessation interventions (pharmacotherapy + behavioral counseling) should be offered to all TB patients who use tobacco; varenicline-based cessation preferred where available	Low → Moderate; Conditional → Strong
TB-alcohol: Alcohol use screening and brief intervention should be offered (Conditional, Low)	TB-Alcohol STAR Trial (Moderate) demonstrates naltrexone + motivational enhancement significantly reduces heavy drinking and improves TB completion	Upgrade: Screening for alcohol use disorder (AUDIT) should be conducted in all TB patients; pharmacotherapy (naltrexone) + motivational enhancement should be offered to patients with harmful alcohol use	Low → Moderate; Conditional → Strong (for screening); Conditional (for pharmacotherapy + motivational enhancement)
TB-malnutrition: Nutritional support should be considered for malnourished TB patients (Conditional, Low)	TB-Malnutrition RCT (Moderate) demonstrates high-energy, high-protein + micronutrient supplementation reduces mortality by 39% and improves treatment success	Upgrade: Nutritional supplementation (high-energy, high-protein, micronutrient-enriched) should be provided to all TB patients with severe malnutrition (BMI <16 adults; weight-for-height <-3 Z-score children) throughout treatment	Low → Moderate; Conditional → Strong

Original Recommendation (Module 6, 2nd ed., 2023)	New Evidence Impact	Updated Recommendation (as of 29 April 2026)	GRADE Quality Change
TB-hepatitis B: No specific recommendation in 2nd ed. (insufficient evidence)	TB-HepB Cohort (Low) demonstrates HBV reactivation risk of 8.2% during isoniazid-containing regimens; prophylactic entecavir reduces reactivation to 1.1%	New recommendation: HBV screening (HBsAg) should be conducted before initiating isoniazid-containing TB treatment or TPT in populations with HBV prevalence ≥2%; prophylactic antiviral therapy (entecavir or tenofovir) should be offered to HBsAg-positive patients starting isoniazid-containing regimens	Low (new); Conditional
TB-silicosis: TPT should be offered to silicosis patients (Conditional, Low)	SilicoTB-Screen Trial (Moderate) demonstrates CAD-enhanced CXR screening detects 34% more TB; targeted counseling increases TPT uptake from 22% to 61%	Upgrade with addition: TPT should be offered to all silicosis patients; annual TB screening using symptom assessment + CAD-enhanced CXR should be implemented for silicosis patients; targeted counseling should accompany TPT offer	Low → Moderate (for screening); Conditional → Strong (for TPT offer); Conditional (for CAD-enhanced screening)
TB-CKD: No specific recommendation in 2nd ed. (insufficient evidence)	TB-CKD Cohort (Very Low) demonstrates 2.3-fold higher mortality; bedaquiline-based regimens show favorable safety in CKD stage 3–4	New conditional recommendation: CKD screening (eGFR) should be considered in TB patients at diagnosis; rifampicin dose adjustment may be required in CKD stage 3–5; bedaquiline-based regimens may be preferred in CKD stage 3–4 for DR-TB	Very Low (new); Conditional
Youth TPT delivery: No specific youth-focused recommendation in Module 6	Youth TPT Acceptability Study (Low) demonstrates youth-informed delivery models significantly increase TPT initiation and completion; shorter regimens (3HP, 1HP) preferred by 89% of youth	New recommendation: Youth-informed TPT delivery models (school-based, peer-navigator, digital adherence support) should be considered for adolescents and young adults (12–24 years); shorter TPT regimens (3HP, 1HP) should be prioritized for youth where eligible	Low (new); Conditional
TB-cancer: No specific recommendation in 2nd ed.	TB-Cancer Registry (Very Low) demonstrates lower treatment success (62%) and higher relapse with immunosuppressive chemotherapy; 4-month regimen shows higher relapse in active cancer	New conditional recommendation: TB patients with active malignancy should receive the standard 6-month DS-TB regimen (4-month regimen not recommended); coordination of TB treatment timing with chemotherapy cycles should be considered; checkpoint inhibitor-associated TB reactivation should be monitored	Very Low (new); Conditional

Summary of GRADE Evidence Quality Shifts

Comorbidity Domain	Previous GRADE (2023)	Updated GRADE (2026)	Direction
TB-DM bidirectional screening	Low	Moderate	↑ Upgrade

Comorbidity Domain	Previous GRADE (2023)	Updated GRADE (2026)	Direction
TB-DM integrated care	Not assessed	Moderate (new)	↑ New
TB-HIV same-day initiation	High (ART); Low (TPT timing)	High (ART); Conditional (same-day TPT+ART)	↔ Maintain + new sub-recommendation
TB-mental health screening	Very Low	Moderate	↑ Upgrade
TB-mental health intervention	Not assessed	Moderate (new)	↑ New
TB-smoking cessation	Low	Moderate	↑ Upgrade
TB-alcohol intervention	Low	Moderate	↑ Upgrade
TB-malnutrition supplementation	Low	Moderate	↑ Upgrade
TB-hepatitis B prophylaxis	Not assessed	Low (new)	↑ New
TB-silicosis screening + TPT	Low	Moderate	↑ Upgrade
TB-CKD management	Not assessed	Very Low (new)	↑ New
Youth TPT delivery	Not assessed	Low (new)	↑ New
TB-cancer management	Not assessed	Very Low (new)	↑ New

This dynamic guideline report was generated following systematic evidence review methodology consistent with WHO GRADE frameworks. All recommendations are subject to further updating as new high-quality evidence emerges. National TB programmes should adapt these recommendations to local epidemiology, resource availability, and health system capacity while maintaining alignment with WHO core principles.